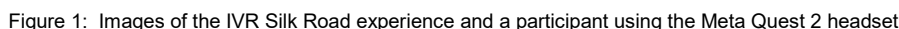


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A recent review of studies using Immersive Virtual Reality (IVR) with children found that IVR contributed to positive learning outcomes [7], however a systematic understanding of how each feature of IVR contributes to its success as a learning experience is still limited. This paper outlines the design of a multisensory embodied IVR experience, with the aim to explore how the inclusion of multisensory and interactive features in IVR contribute to the pedagogical success of the experience.

1 BACKGROUND

This workshop paper outlines the development and pilot of an educational IVR world with the aim to assess the impact of the multisensory (in particular: auditory and haptic) and interactive elements of IVR on learning outcomes.

To assess the impact of IVR on learning outcomes, two key measures were taken after the experience: factual knowledge recall and engagement. Factual knowledge recall was chosen as a useful measure to assess the effectiveness of IVR, as there is disagreement in the literature as to whether or not IVR is beneficial for learning facts about a topic. Recent research has found that teaching topics such as engineering [2], biochemistry [18] and neuroanatomy [10] in IVR led to significantly higher factual knowledge recall, compared to teaching the same topics through more traditional teaching methods. However, other studies have found less interactive and immersive educational experiences to be more effective at promoting factual knowledge recall than IVR [21]. As retaining factual knowledge is a key metric of successful learning [6], it is important to assess how useful IVR is at this.

Furthermore, IVR has been reported to be more engaging than traditional teaching methods across multiple studies [7]. For example, students reported higher engagement in an IVR biochemistry learning experience compared to a video-based lesson [4]. Engagement is linked to learning outcomes and general student success [29], so it is something to be prioritised when designing any new educational tool. However, there is a lack of evidence to suggest which multisensory and interactive elements in IVR are driving this high engagement, so by systematically testing each of these elements we hope to shed some light here.

In particular, this paper will focus on exploring how children aged between 7- to 13-years-old learn from a multisensory embodied IVR experience. In the existing literature the majority of research on educational IVR explores IVR use in older children and young adults [30], however, recent statistics have shown that children as young as 6 are using VR headsets at home [3] as household ownership of VR headsets increases [24]. It can be expected that younger children will experience IVR differently, as

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their integration of audio and visual information is different to adults [16], so exploring the experience of younger children in IVR is vital in designing educational IVR that is suitable for all.

Studies assessing the utility of educational IVR often measure feelings of embodiment and presence in IVR, as these two affordances of IVR are thought to increase engagement and task performance [19]. Embodiment is the sense of agency and control over a virtual body in IVR and presence is the sense of “being there” in IVR. There is evidence to suggest that higher presence in an IVR learning experience can lead to improved learning outcomes [22], however it is unclear if certain elements of the multisensory IVR experiences are driving this higher presence more than others, therefore this research aims to explore whether differences in the sense of presence in IVR affect the pedagogical success of the IVR experience.

Finally, the majority of studies interested in how IVR can be used as an educational tool don’t consider how individual differences between participants might be impacting how much they learn from the experience. This is especially important to consider when designing an IVR experience for children, as sensory environments, such as IVR experiences, can be experienced differently by children who have hyper- or hypo-sensory sensitivity [23], which may impact their learning experience in an audio-visual IVR experience. Furthermore, younger children have poorer audio-visual integration abilities compared to adults [5, 13]. This poorer integration of audio and visual information in younger children may increase the cognitive load experienced in an IVR experience and subsequently reduce their ability to learn from the experience. Therefore, individual differences in sensory sensitivity and audio-visual integration abilities will be measured.

This paper reports the design of an educational IVR experience, designed to teach 7- to 13-year-old children. It will allow us to answer major research questions such as (RQ1) how is the pedagogical success of the IVR experience influenced by the age of participants and the inclusion or exclusion of auditory stimuli or interactive and haptic elements? Next (RQ2) we will ask how individual differences in the sense of presence in IVR affect the pedagogical success of the IVR experience. Finally, we will ask (RQ3) how the learning outcomes from an IVR experience, the unique affordances of IVR, and the individual differences between users relate to each other.

2 DESIGNING THE EDUCATIONAL IVR EXPERIENCE

Evidence from existing systematic reviews suggest that the majority of experimental research so far has focused on how IVR can be used as a teaching tool in science education [12, 30], from climate change [22] to the phases of the moon [20], with a clear lack of research on using IVR in the humanities, in particular: History. Whilst History VR experiences are being produced, they are generally created with the aim to recreate a Historical location for the benefit of either History academics attempting visualise their time period of interest [9] or for tourism purposes [26], rather than the aim to develop an experimentally backed learning experience for students in the classroom. The current paradigm will allow us to fill this gap in the literature.

We chose to base our educational experience around a Historical time period, to utilise IVR’s unique ability to virtually transport users through time and space, so they can experience historical moments “first hand”. Furthermore, History can be a subject that is difficult to teach in schools: studies suggest UK school children struggle to articulate why History is a useful subject to learn in school [15] and there is a preference for more interactive History lessons [14].

For the current experiment four Historical time periods were short-listed to be the basis of our educational experience: the Empire of Alexander the Great; the Mughal Empire; the Silk Road; and Ancient Sumer and the development of the first writing system. The Silk Road was ultimately chosen, as setting the experience in one location, a market place on the Silk Road, allowed us to display a large number of artefacts and objects from the time period in a more realistic way. Furthermore, the Silk Road was an excellent example of cultural exchange and the beginning of globalisation, which is a pertinent topic today. It was decided to choose a historical topic not in the UK History curriculum [6] to ensure the participants would have no to minimal prior knowledge of the topic. Finally, setting the experience on the Silk Road, specifically in Ancient China, allowed us to take advantage of the unique feature of IVR that allows participants to visit locations far removed from the classroom, that are not easily accessible on a school trip.

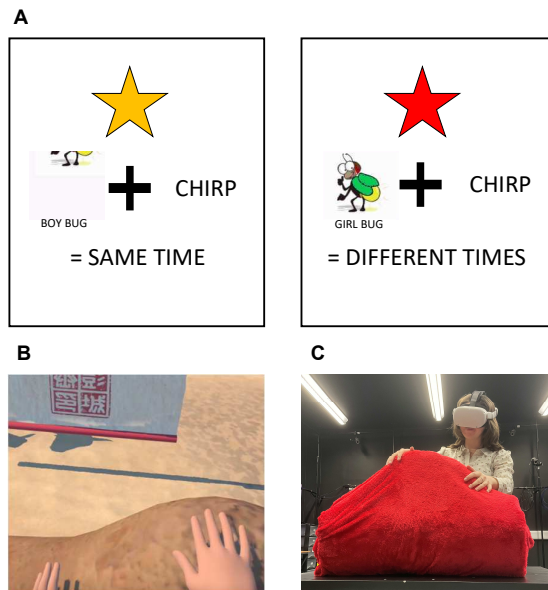


Figure 2: Participant instructions for the flash-beep task (A); the virtual hands used in the Pilot Study 2 Hand and Haptic conditions (B); the camel hump used to produce a passive haptic sensation in the Pilot Study 2 Haptic condition (C)

The IVR experience was custom-built in Unity and was set within a desert scene, which included: sand dunes and mountains in the distance; an array of goods and artefacts on market stalls (e.g., rolls of silk, vases, fruits) and animated animals (camels and sheep). Participants were able to walk around within a 2m x 2m square area to closely explore the items on the market stalls. The hardware used in this study was a Meta Quest 2, with external headphones attached and the participants interacted with the world either with Meta Quest 2 controllers or virtual tracked hands. The experience was narrated and participants followed instructions to select objects in the IVR world, which, when selected, prompted a Chinese hanging scroll to unravel and reveal facts about the object.

3 METHODS

3.1 Pilot study 1

This consisted of two age groups of children: 7- to 9-year-olds and 11- to 13-year-olds. Participants were also assigned to one of two SoundState conditions (SoundOn or SoundOff). Both the SoundOn and SoundOff conditions included a voiceover narration, but only the SoundOn condition included background Chinese-style music, the sounds of animals, and user interface sounds; the SoundOff condition did not include these elements.

The pedagogical success of the IVR experience was measured first through a custom factual knowledge recall quiz (which included questions such as: “How long can camels survive in the desert without water?”). The pedagogical success of the IVR experience was also measured through a 6-item engagement questionnaire adapted from the Museum Experience Scale [25]. An example question from the engagement questionnaire: “Did you enjoy your experience in the VR world?”.

Two affordances of IVR were also measured using established questionnaires: presence, using the established Slater-Usuh-Steed (SUS) Presence Questionnaire [27] and embodiment, using the standardised embodiment questionnaire developed by Gonzalez-Franco & Peck [11].

Two measures of individual differences were included. The first was the audio-visual temporal binding window (how temporally close audio and visual stimuli must be in order to be perceived as simultaneous [16]). This was measured using a flash-beep task, in which a participant has to determine whether or not an auditory and visual stimuli occur simultaneously. The flash-beep task was adapted from a published protocol designed to measure audio-visual temporal binding windows in children from 6-years-old [16], see Figure 2a for the participant instructions for this flash-beep task. The second measure of individual differences was sensory sensitivity (measured using the adapted Glasgow Sensory Questionnaire [23]).

3.2 Pilot study 2

Participants aged 7-11 years were assigned to one of three Interactivity State conditions. In the Look condition participants did not have virtual hands and instead selected objects in the virtual world using gaze direction (i.e., looking at objects). In the Hand and Haptic conditions, participants selected objects using tracked virtual hands (see Figure 2b), with the Haptic condition adding the element of passive haptics. This was achieved by placing objects in the real world such that when the virtual object (e.g. camel; pots) was touched by the virtual hand, the participant’s real hand touches the real object (see Figure 2c). The pedagogical success and affordances of IVR were measured in the same way as in the first pilot study. The temporal binding window was not measured here, but sensory sensitivity was measured as above.

4 PRELIMINARY FINDINGS

4.1 Pilot study 1

From a pilot study with 10 participants, we found no significant difference in factual recall or engagement between the SoundOn and SoundOff conditions (RQ1), however there were high factual recall and engagement scores across all conditions. It was, however, found that the level of presence in the IVR experience impacted the relationship between age group and engagement (RQ2). There was a correlation between individual differences in audio-visual temporal binding windows and factual recall for the younger children (RQ3).

4.2 Pilot study 2

From a pilot study with 12 participants, we found no significant difference in factual recall between the Look, Hand and Haptic conditions, however there were high factual recall scores across all conditions. However, we found higher engagement scores in the high interactivity condition (Hand) than the low interactivity condition (Look). Adding passive haptics (Haptic condition) did not significantly increase engagement scores compared to the Hand condition (RQ1). The level of presence did not impact the relationship between Interactivity State and the learning outcomes (RQ2).

5 CONCLUSION

The main aim of this research was to create an educational virtual experience that can be systematically adjusted to answer these three research questions: (RQ1) how is the pedagogical success of the IVR experience influenced by the age of participants and the inclusion or exclusion of auditory stimuli or interactive and haptic elements; (RQ2) how do individual differences in the sense of presence in IVR affect the pedagogical success of the IVR experience; and (RQ3) how do the learning outcomes from an IVR experience, the unique affordances of IVR, and the individual differences between users relate to each other? The pilot studies conducted here suggests that we have created a viable environment in which to test these questions. Following on from these pilot studies we intend to collect a full sample of participants to fully test and answer these research questions.

This research also illustrates the potential utility of embodied multisensory IVR as an educational tool and outlines how future VR developers could go about designing IVR experiences for an applied educational setting.

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